Tut. 6 – Solutions

DC 427

Tute 6 soln

1. The characteristic equation of the closed loop system is

$$\det(zI - (\Phi - HL)) = z^2 - (a_{11} + a_{22} - b_2\ell_2 - b_1\ell_1)z + a_{11}a_{22} - a_{12}a_{21} + (a_{12}b_2 - a_{22}b_1)\ell_1 + (a_{21}b_1 - a_{11}b_2)\ell_2$$

Identifying with the desired characteristic equation gives

$$\begin{pmatrix} b_1 & b_2 \\ a_{12}b_2 - a_{22}b_1 & a_{21}b_1 - a_{11}b_2 \end{pmatrix} \begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} = \begin{pmatrix} p_1 + \operatorname{tr}\Phi \\ p_2 - \det\Phi \end{pmatrix}$$

where $\operatorname{tr}\Phi = a_{11} + a_{22}$ and $\det\Phi = a_{11}a_{22} - a_{12}a_{21}$. The solution is

$$\begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} = \frac{1}{\Delta} \begin{pmatrix} a_{21}b_1 - a_{11}b_2 & -b_2 \\ -a_{12}b_2 + a_{22}b_1 & b_1 \end{pmatrix} \begin{pmatrix} p_1 + \operatorname{tr}\Phi \\ p_2 - \det\Phi \end{pmatrix}$$

where

$$\Delta = a_{21}b_1^2 - a_{12}b_2^2 + b_1b_2(a_{22} - a_{11})$$

To check when $\Delta \neq 0$ we form the controllability matrix of the system

$$W_c = \begin{pmatrix} H & \Phi H \end{pmatrix} = \begin{pmatrix} b_1 & a_{11}b_1 + a_{12}b_2 \\ b_2 & a_{21}b_1 + a_{22}b_2 \end{pmatrix}$$

and we find that

$$\Delta = \det W_c$$

There exists a solution to the system of equations above if the system is controllable.

2. In this case the desired characteristic equation is

$$(z - 0.1)(z - 0.25) = z^2 - 0.35z + 0.025.$$

Using the result from Problem 1 we find that

$$\Delta = 0.5$$

and L is obtained from

$$L^T = \begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} = \frac{1}{\Delta} \begin{pmatrix} 0.5 & 0 \\ 0.1 & 1 \end{pmatrix} \begin{pmatrix} 0.75 \\ -0.025 \end{pmatrix} = \begin{pmatrix} 0.75 \\ 0.1 \end{pmatrix}$$

To check the result we form

$$\Phi - HL = \begin{pmatrix} 1 & 0.1 \\ 0.5 & 0.1 \end{pmatrix} - \begin{pmatrix} 0.75 & 0.1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0.25 & 0 \\ 0.5 & 0.1 \end{pmatrix}$$

The matrix $\Phi - HL$ thus has the desired eigenvalues 0.25 and 0.1.

3. Using the results in Problem 4.1 gives

$$L^T = \frac{1}{-0.0029}\begin{pmatrix} -0.0880 & -0.1600 \\ -0.0125 & 0.0100 \end{pmatrix}\begin{pmatrix} 0.5900 \\ -0.1595 \end{pmatrix} = \begin{pmatrix} 9.22 \\ 3.11 \end{pmatrix}$$

4. a. In this case

$$W_o = \begin{pmatrix} C \\ C\Phi \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0.22 & 1 \end{pmatrix}, \quad W_o^{-1} = \begin{pmatrix} -4.55 & 4.55 \\ 1 & 0 \end{pmatrix}$$

$$\Omega = \begin{pmatrix} 0 \\ C\Gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 0.03 \end{pmatrix},$$

$$\Psi = \Gamma - \Phi W_o^{-1}\Omega$$

$$= \begin{pmatrix} 0.22 \\ 0.03 \end{pmatrix} - \begin{pmatrix} 0.78 & 0 \\ 0.22 & 1 \end{pmatrix}\begin{pmatrix} -4.55 & 4.55 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 0.03 \end{pmatrix} = \begin{pmatrix} 0.114 \\ 0 \end{pmatrix}$$

We get

$$\hat{x}(k) = \Phi W_o^{-1}\begin{pmatrix} y(k-1) \\ y(k) \end{pmatrix} + \Psi u(k-1)$$

$$= \begin{pmatrix} -3.55 & 3.55 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} y(k-1) \\ y(k) \end{pmatrix} + \begin{pmatrix} 0.114 \\ 0 \end{pmatrix} u(k-1)$$

b. The dynamical observer (4.28) has the form

$$\hat{x}(k+1|k) = (\Phi - KC)\hat{x}(k|k-1) + \Gamma u(k) + Ky(k).$$

In this case we choose K such that the eigenvalues of $\Phi - KC$ are in the origin. Using the results from Problem 4.1 but with Φ^T and C^T instead of Φ and Γ gives

$$K = \begin{pmatrix} 2.77 \\ 1.78 \end{pmatrix}$$

Solutions of this tutorial were taken from Astrom & Wittemark, "Solution manual of Computer-controlled Systems," Prentice Hall, 1997.